Intelligent Vehicle Recommendation System

Using Random Forest, CART, and Cosine Similarity

Adithyan R (305) Aadil Sandeep (301) Adhway Satheesh (366)

SCTCE College
Project Guide: Prof. Rejimoan .R

March 17, 2026

Table of Contents

Introduction

Project Overview & Purpose

- **AutoMatch:** An intelligent vehicle recommendation system.
- **Purpose:** Translates user lifestyle habits and budget constraints into technical vehicle specifications to provide highly personalized recommendations.

Project Objectives

- Build a predictive and robust classification model using **Random Forest** and **CART**.
- Implement **Cosine Similarity** for content-based ranking.
- Provide a user-friendly, interactive web interface using **Streamlit**.

Problem Statement

The Challenge

- Navigating vast automotive catalogs is overwhelming for standard users without technical knowledge.
- Traditional platforms rely on manual hard-filters (e.g., body type, exact price) rather than understanding "lifestyle" requirements (e.g., family trips, off-roading).

Our Goal

- Develop an intelligent system that predicts preferences and suggests vehicles based on technical specs and user similarity.
- Seamlessly map human lifestyle choices to complex vehicle engineering parameters.

Methodology: A Two-Stage Process

Stage 1: Classification (Filtering)

- *Random Forest (RF):* Primary model for high-accuracy classification of cars into budget buckets (Low, Mid, High). Robust and prevents overfitting.
- *CART (Decision Tree):* Used alongside RF for explainability. Defines clear logical rules for why a car belongs to a certain class.

Stage 2: Ranking (Personalization)

- *Cosine Similarity:* Calculates the mathematical "angle" between what the user wants and what the car offers.
- Higher value (closer to 1) means the car's engine, mileage, and utility perfectly align with the user's lifestyle:

$$\text{similarity} = \cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$

Implementation Design

Data Preprocessing

- Text Extraction Currency Cleaning (Regex Pandas)
- Missing Value Handling (Median Imputation Fallback Logic)
- Feature Engineering Discretization (Quantile Binning)

Modular Architecture Design

- **Frontend:** Streamlit-based UI Layer & Input Handling.
- **Logic & ML Layer:** Orchestration of filtering/ranking and ML Model definitions.
- **Lifestyle-to-Specs Mapping:** Automatically converts inputs to requirements (e.g., "Family Trips" → high seating capacity).

Results & Current Status

Achievements

- Successfully integrated trained Random Forest and CART models with the Streamlit web interface.
- Deployed a robust recommendation engine that efficiently filters and personalizes car suggestions.
- Added an *Explainability Feature* — every recommendation includes a "Why this car?" snippet to build user trust.